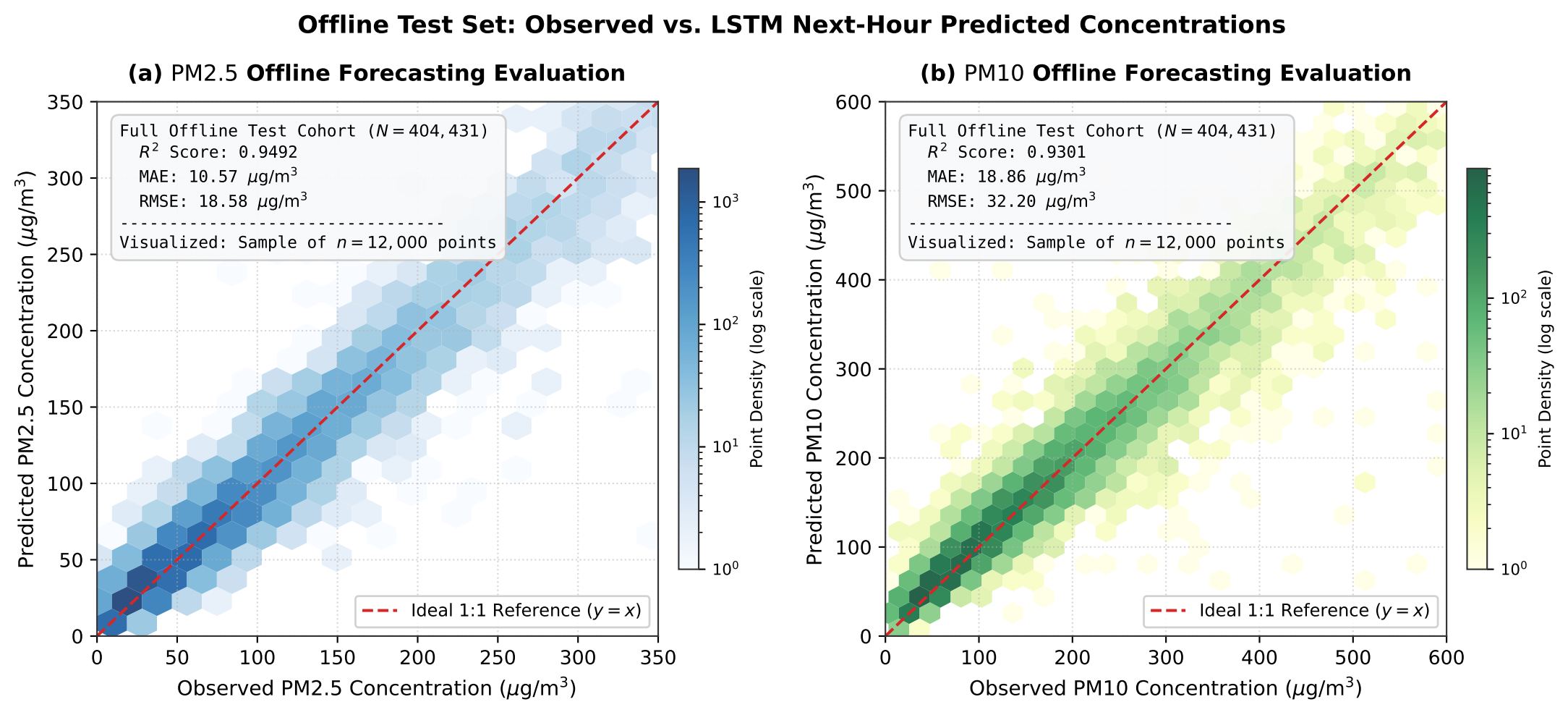

Offline Test Set: Observed vs. LSTM Next-Hour Predicted Concentrations
(a) PM2.5 Offline Forecasting Evaluation
Full Offline Test Cohort (N = 404,431)
R² Score: 0.9492
MAE: 10.57 μg/m³
RMSE: 18.58 μg/m³
Visualized: Sample of n = 12,000 points
Ideal 1:1 Reference (y = x)
Predicted PM2.5 Concentration (μg/m³)
Observed PM2.5 Concentration (μg/m³)
Point Density (log scale)
(b) PM10 Offline Forecasting Evaluation
Full Offline Test Cohort (N = 404,431)
R² Score: 0.9301
MAE: 18.86 μg/m³
RMSE: 32.20 μg/m³
Visualized: Sample of n = 12,000 points
Ideal 1:1 Reference (y = x)
Predicted PM10 Concentration (μg/m³)
Observed PM10 Concentration (μg/m³)
Point Density (log scale)